AI MINEGUARD

SIH26024 — AI-Based Smart Governance and Compliance Monitoring System for Coal Mines

MODULE 2

Field Inspector & Mobile Reporting

Functional Design, Field Operations, Offline Workflow & Demonstration Documentation

Version 1.0 — Implementation Baseline

Field Reporting • Offline Support • GIS-ready Geo-tagging • Inspection Management • Incident Management • Field Verification

1. Document Control

Item	Details
Document status	Implementation baseline / functional design
Module	Module 2 — Field Inspector & Mobile Reporting
Depends on	Module 1 — AI Compliance & Risk Engine; Phase 1 Auth/IAM and Phase 2 Mine/Org hierarchy
Primary users	Field Inspector, Safety Officer, Environment Officer, Mine Manager, authorized verifier
Primary outcome	Capture trustworthy field observations even without connectivity and convert them into traceable inspections, incidents, violations and verification workflows
Core principle	Capture once at the field; preserve evidence, location, time, author and synchronization state; verify before official closure

2. Executive Summary

Module 2 is the field execution layer of AI MineGuard. Module 1 provides automated sensing and risk reasoning, but inspectors still need to physically inspect mine conditions, collect evidence and report what they observe. The supplied roadmap defines Module 2 as **Field Inspector & Mobile Reporting**, with **geo-tagged offline inspections** and **AI-assisted incident structuring** as its core deliverables.

The mobile application should function as a **digital field notebook + evidence collector + inspection workflow client**. It must remain useful in weak or zero-connectivity areas. An offline report is a durable local record with a synchronization lifecycle, not a disposable draft.

> The governance model follows Module 1: AI can assist perception or structuring, but official compliance decisions remain controlled by deterministic rules and authorized human verification.

3. Why Module 2 Exists

Field problem	Why Module 2 solves it	Result
Remote mine areas	Mobile-first reporting with offline storage	No continuous internet dependency
Paper/manual notes	Structured forms and required fields	Consistent records
Uncertain observation location	GPS + timestamp + mine/zone	Geo-traceable evidence
Scattered media	Photo/video/voice linked to report	Single evidence package
Fast incident response needed	Dedicated incident workflow	Faster escalation
False/unchecked closure risk	Verification state machine	Evidence-based closure
AI signal needs field validation	Create inspection/verification work from upstream signals	AI + human validation

4. Module 2 at a Glance

> **Mobile App → Local DB/Queue → Sync Service → Central API → Module 1 / GIS / Dashboard → Verification & Corrective Action**

1. **Field Reporting:** capture mine, location, GPS, timestamp, officer, observation and evidence.
2. **Offline Support:** continue creating reports with no internet and synchronize later.
3. **GIS Mapping:** send trustworthy coordinates and map-ready field features.
4. **Inspection Management:** plan, schedule, assign, execute and submit inspections.
5. **Incident Management:** record accidents, injuries, fire, equipment failure, environmental and safety incidents.
6. **Field Verification:** move issues from reported through verified, corrective action, evidence, review and closure.

5. Feature Catalogue

ID	Feature	What it does	Primary output
F2-01	Field Report	Creates a geo-tagged observation with evidence	Field report
F2-02	GPS Capture	Captures latitude, longitude, accuracy and time	Geo-coordinate
F2-03	Evidence Capture	Captures photo, video, voice and text	Evidence items
F2-04	Offline Submit	Stores report locally when network is unavailable	Queued report
F2-05	Auto Sync	Uploads queued records when connectivity returns	Synced report
F2-06	Sync Status	Shows pending, syncing, synced, failed, conflict	Sync state
F2-07	Inspection Templates	Digital checklists by inspection type	Checklist
F2-08	Inspection Scheduling	Assigns inspector/date/location	Inspection task
F2-09	Incident Report	Records structured incident details	Incident
F2-10	AI Incident Structuring	Suggests structured fields from voice/text	AI suggestions
F2-11	GIS Handoff	Feeds coordinates and incident metadata	Map feature
F2-12	Field Verification	Tracks verification and remediation	Verified/closed case
F2-13	Evidence Timeline	Chronological evidence/action history	Audit timeline
F2-14	Notifications	Assignments, failed syncs, review requests	Notification

6. Field Reporting — What the Inspector Does

The officer selects the mine and context, starts a report, and captures the observation. The app should automatically collect physical context wherever possible.

Minimum report payload:

- Mine ID and optional zone ID
- Inspector/officer ID
- Report type and observation category
- GPS latitude, longitude and accuracy
- Physical capture timestamp
- Text observation/description
- Severity or urgency where applicable
- Photos and/or video
- Optional voice note
- Related inspection, incident, violation or Module 1 event
- Device/app and synchronization metadata

Example: an inspector observes a safety issue near a haul road. They capture a photo, add a voice note, record GPS and submit. The record remains traceable because the system stores who captured it, where, when and with what evidence.

7. Offline Support — Critical Design

Mining sites can have dead zones or unstable networks. Offline mode is therefore a first-class operating state.

NO INTERNET → CREATE REPORT → LOCAL VALIDATION → LOCAL DB → SYNC_PENDING → INTERNET RETURNS → SYNC QUEUE → CENTRAL SERVER → ACKNOWLEDGEMENT

State	Meaning	Inspector action
DRAFT	Started but not submitted	Continue editing
READY_TO_SYNC	Validated and waiting	No action required
SYNCING	Upload in progress	Wait / background sync
SYNCED	Server acknowledged	Normal
SYNC_FAILED	Upload failed	Retry; retain local copy
CONFLICT	Versions differ	Resolve according to policy

Offline data should be encrypted at rest on the device. The local queue should use idempotency keys so a retry cannot create duplicate central reports.

8. GIS Mapping — Why It Matters

Module 2 is not the complete GIS module; the roadmap assigns advanced GIS and satellite intelligence to Module 4. Module 2's job is to produce reliable field coordinates and map-ready features.

Map item	Module 2 contribution	Future consumer
Mine	Selected mine context	GIS/dashboard
Inspection location	GPS point + inspection ID	GIS
Incident location	GPS + type/severity	GIS
Violation location	Coordinates + governance state	GIS/risk engine
High-risk area	Field evidence context	Module 1 + GIS
Visit trail	Optional visit points	GIS/analytics

Location quality rules

- Capture latitude/longitude automatically when permission is available.
- Store GPS accuracy; do not imply precision that the device did not provide.
- Record physical event time separately from server ingestion time.
- Resolve coordinates to mine/zone where possible.
- Allow manual location only when GPS is unavailable and flag it.
- Do not silently overwrite captured coordinates during synchronization.

9. Inspection Management

Inspection Management converts a field visit into a repeatable process: **planning** → **execution** → **review**.

Planning:

- Choose Fire Safety, Environmental, Statutory, Regulatory or configurable inspection type.
- Choose mine/zone and date/time.
- Assign an authorized inspector.
- Attach a checklist/template and optional prior findings.

Execution:

- Open assigned inspection.
- Work through checklist items.
- For failed/unsafe items capture observation and evidence.
- Capture GPS and create incident directly from a failed item.
- Submit even when offline.

Review:

- Supervisor reviews submission.
- Verified issues become violations/corrective actions as appropriate.
- Retain evidence and review decision.
- Close only through authorized workflow.

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PLANNED → ASSIGNED → IN_PROGRESS → SUBMITTED → UNDER_REVIEW → VERIFIED /
REQUIRES_ACTION → CLOSED
```

10. Incident Management

An incident is a time-sensitive event requiring response, investigation and potentially corrective action. The incident form should be faster than a generic report.

Supported types

- Accident
- Injury
- Fire
- Equipment failure
- Environmental incident
- Safety incident
- Other configurable type

Minimum incident record

Field	Purpose
Incident ID	Unique traceable identifier
Incident type	Classifies the event
Date/time	Physical occurrence time
Location	GPS + mine/zone
Reporter	Officer who captured it
People involved	Affected/involved people where operationally appropriate
Witnesses	Witness information where policy permits
Severity	Routing/urgency
Description	Human-readable event description
Evidence	Photo/video/audio/document references
Immediate action	What was done at site
Status	Reported / Verified / Action Required / Closed

11. AI-Assisted Incident Structuring

The roadmap specifically calls for AI-assisted incident structuring. Implement this as **assistive, not autonomous**. The officer's voice/text remains source evidence; AI suggests structured fields that the officer can edit and confirm.

Raw voice note: “At the north haul road around 10:20, the dumper stopped suddenly and smoke was noticed near the engine compartment. No injury observed.”
AI suggestions: incident_type = EQUIPMENT_FAILURE; possible_hazard = SMOKE; location_context = NORTH_HAUL_ROAD; occurrence_time = 10:20; injury = NONE_REPORTED. Officer reviews and confirms.

AI governance rules

- Keep original voice/text unchanged.
- Store AI suggestions separately from confirmed fields.
- Record model name/version and inference time.
- Never silently overwrite officer-entered values.
- Require officer confirmation for severity and official compliance fields.

12. Field Verification Workflow

REPORTED → VERIFIED → CORRECTIVE ACTION → EVIDENCE SUBMITTED → SUPERVISOR REVIEW → APPROVED → CLOSED

Stage	What happens	Who normally acts
REPORTED	Field officer submits observation/incident	Inspector/officer
VERIFIED	Authorized reviewer confirms issue	Safety/Environment/authorized verifier
CORRECTIVE ACTION	Remediation task is created	Manager/responsible owner
EVIDENCE SUBMITTED	Fix evidence is uploaded	Owner/contractor
SUPERVISOR REVIEW	Evidence checked against issue	Supervisor/inspector
APPROVED	Remediation accepted	Authorized verifier
CLOSED	Case formally closed	System + authorized user

13. How Module 2 Connects to Module 1

Module 2 event	Downstream action
Field safety observation	Candidate/verified violation or safety signal
Failed inspection item	Violation/corrective-action candidate
Environmental field reading	Environmental risk input
Equipment inspection finding	Equipment compliance/corrective action
Incident severity	Alert/risk prioritization input subject to rules
Repeated field issue	Historical/recurrence signal
Verified field violation	Risk score recomputation

Module 2 should not become a second risk engine. It captures facts; the common governance/rule layer determines how those facts affect official compliance and risk.

14. Roles & Permissions

Role	Responsibility	Key permissions
Inspector	Execute inspections, reports, remediation verification	field.report.create, inspection.execute, action.verify
Safety Officer	Review safety incidents/violations	incident.read, violation.verify, inspection.review
Environment Officer	Environmental inspections/readings	environment.capture, inspection.review
Mine Manager	Assign and monitor work	inspection.assign, action.manage, report.read
Corporate Admin	Cross-mine governance	cross_mine.read, report.read
Contractor	Assigned corrective actions	action.read_assigned, evidence.upload
Viewer	Read-only visibility	report.read

15. Security & Trust Principles

- Reuse platform authentication and mine context.
- Enforce role-based permissions on every workflow transition.
- Do not trust client-provided tenant/mine IDs without authorization.
- Protect evidence with immutable/versioned references.
- Audit creation, edits, submission, verification, synchronization and closure.
- Encrypt offline data and support device revocation.
- Prevent cross-mine exposure during synchronization.

16. Testing Strategy

Test layer	What to test
Unit	GPS validation, state transitions, severity rules, idempotency
Mobile	Offline create/edit/submit, attachments, permissions, local security
Sync	Retry, duplicate prevention, partial upload, acknowledgement, conflicts
API	Auth, authorization, payload validation, evidence upload, CRUD
Workflow	Reported → verified → action → evidence → review → closed
GIS	Coordinate accuracy, zone resolution, manual-location flag

Test layer	What to test
AI	Structuring quality, schema validity, confirmation flow
Security	Cross-mine access, revoked users, offline-device scenarios
Performance	Large media queues, batch sync, weak-network behavior

17. Acceptance Criteria

1. Inspector can create a field report with officer, timestamp, GPS and observation.
2. Photo, video and voice evidence can be attached.
3. Report can be created and submitted with zero connectivity.
4. Offline report survives app restart and remains in sync queue.
5. Connectivity restoration synchronizes without duplicates.
6. Original capture time and coordinates are preserved.
7. Inspection can be scheduled, assigned, executed and submitted.
8. Incident can be created with type, severity, location, description and evidence.
9. AI suggestions never overwrite original officer input.
10. Reported issue cannot be closed without configured verification.
11. Corrective evidence can be uploaded and reviewed.
12. Important transitions are auditable.
13. Records are available to downstream GIS and Module 1 through versioned APIs.

18. Recommended MVP — What Your Team Should Actually Build

For a hackathon, build one strong vertical slice rather than every possible feature.

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LOGIN → SELECT MINE → START INSPECTION → GPS → CHECKLIST → PHOTO → VOICE/TEXT → SUBMIT
OFFLINE → SYNC PENDING → RECONNECT → AUTO SYNC → CENTRAL RECORD → MAP POINT →
REVIEW → CORRECTIVE ACTION → AFTER-FIX EVIDENCE → VERIFY → CLOSE
```

Priority	Build	Why
P0	Login + mine selection	Identity/scope
P0	Field report form	Core value
P0	GPS + timestamp	Trustworthy evidence
P0	Photo capture	Visible proof
P0	Local offline storage	Mining differentiator
P0	Sync queue + retry	Completes offline promise
P0	Inspection checklist	Structured inspection
P0	Incident creation	Incident path
P0	Verification + corrective action	Governance loop
P1	Voice + AI structuring	Roadmap differentiator
P1	GIS map screen	Visible geo-tagging
P2	Advanced analytics/tracking	Later enhancement

19. Demonstration Script

1. Open mobile app as Field Inspector.
2. Select Mine A and start a Fire Safety inspection.
3. Show automatic GPS and timestamp.
4. Fail a checklist item and capture a photo plus voice/text.

5. Disable network.
6. Submit and show SYNC_PENDING instead of failure.
7. Reconnect and show automatic synchronization.
8. Open central view and show the same evidence and map location.
9. Verify issue and create corrective action.
10. Upload after-fix evidence.
11. Supervisor verifies and closes.
12. Open audit history.

20. Final Module 2 Principle

Capture in the field → preserve evidence → work offline → synchronize reliably → map the observation → verify the issue → remediate → review evidence → close → audit

Reference basis: the supplied Module 1 documents define Module 2 as Field Inspector & Mobile Reporting with geo-tagged offline inspections and AI-assisted incident structuring, and establish the platform's human-verification, evidence and audit principles.